

VIPER-SOC

Vulnerable IP & Payload Eradication Response

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1. Executive Summary

VIPER (Vulnerable IP & Payload Eradication Response) is an automated, dual-layer incident response framework integrated with Wazuh SIEM. Built to eliminate high MTTR during active intrusions, VIPER pairs real-time threat intelligence (AbuseIPDB & VirusTotal) with host-level Active Response daemons. It delivers zero-touch, millisecond containment by dynamically blocking malicious IPs at the network boundary and instantly purging payloads from the filesystem.

2. Threat Landscape & Problem Statement

Modern Security Operations Centers (SOCs) face two primary operational bottlenecks:

- **Slow Manual Triage:** Manual verification of compromised endpoints or malicious inbound traffic through external threat lookup tools creates a dangerous latency window (often minutes to hours) where an attacker can lateralize or execute secondary payloads.
- **Incomplete Protection:** Security controls often isolate either the host or network traffic, leaving open vectors. VIPER addresses this by establishing Dual-Layer Correlation, simultaneously severing network communication and removing filesystem threats.

2. Architecture Setup

Pillar 1: Network Defense Setup (AbuseIPDB Integration)

Wazuh Manager Configuration: Deployed a custom Python executable hook (`/var/ossec/integrations/custom-abuseipdb`) on the manager. When network connection logs or auth failures are parsed, the manager extracts target remote IP addresses and queries the AbuseIPDB REST API.

Agent Active Response Execution: Upon parsing an IP confidence score $\geq 50\%$ (Rule 100101), the manager transmits an Active Response command to the local agent's active-response/bin/host-deny script, immediately writing the malicious IP into `/etc/hosts.deny`.

Verification Command (IP Block Test): Simulated brute-force SSH authentication failures to test real-time IP enrichment and host-deny blocking:

```
logger -t sshd "Failed password for invalid user hacker from 66.116.242.235 port 22 ssh2"
```

Pillar 2: Endpoint File Defense (VirusTotal Integration)

File Integrity Monitoring (FIM): Configured Wazuh's syscheck module on target directories (e.g., /tmp/, /var/tmp/, /home/user/downloads/) to run real-time hash checks (SHA256) upon file creation or modification.

Malware Eradication Script: Configured the manager's internal virustotal.py module to submit new file hashes to the VirusTotal API. When positive malicious detections are confirmed (> 0), Rule 100200 triggers an active response calling `rm -f` directly on the payload path on the agent endpoint.

Verification Command (FIM Malware Sample - Safe): Injected a non-malicious EICAR test string into a monitored directory to validate real-time hash hashing:

```
echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > /tmp/eicar.bin
```

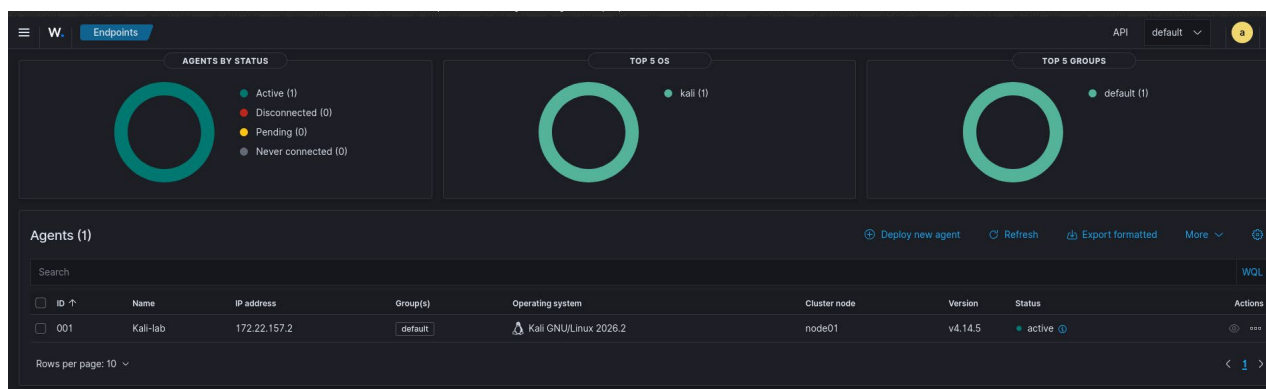


Figure 1 : Active Linux agent connected to Wazuh Manager with Active Response enabled.

```
<!-- AbuseIPDB Base Integration Rule -->
<rule id="100100" level="3">
  <field name="integration">custom-abuseipdb</field>
  <description>AbuseIPDB lookup executed.</description>
</rule>

<!-- AbuseIPDB High Abuse Score Alert Rule -->
<rule id="100101" level="10">
  <if_sid>100100</if_sid>
  <field name="abuseipdb.abuseConfidenceScore" type="pcre2">^([5-9][0-9]|100)$</field>
  <description>AbuseIPDB: High Abuse Confidence Score detected!</description>
  <group>threat_intel,abuseipdb,</group>
</rule>

<!-- VirusTotal High Severity Alert Rule -->
<rule id="100200" level="12">
  <if_sid>87105</if_sid>
  <description>VirusTotal: Alert - File matches known malware signature!</description>
  <group>virus,malware,threat_intel,virustotal,</group>
</rule>
</group>
```

Figure 2 : Custom Wazuh detection rules evaluating AbuseIPDB confidence scores and VirusTotal threat signatures.

3. Core Capabilities & Defense Modules

Module 1: Network Defense (AbuseIPDB Integration)

- **Monitored Vector:** Inbound and outbound IP connections captured by system logs.
- **Threat Intelligence:** Queries the AbuseIPDB API in real time to fetch IP reputation metrics.
- **Trigger Condition:** Confidence score exceeds 50%.
- **Automated Remediation:** Generates a Level 10 Alert on the manager and automatically appends the offending IP to /etc/hosts.deny on the target agent.

```
(jkr@JKR)-[~]
$ sudo cat /etc/hosts.deny
# /etc/hosts.deny: list of hosts that are _not_ allowed to access the system.
# See the manual pages hosts_access(5) and hosts_options(5).
#
# Example:      ALL: some.host.name, .some.domain
#              ALL EXCEPT in.fingerd: other.host.name, .other.domain
#
# If you're going to protect the portmapper use the name "rpcbind" for the
# daemon name. See rpcbind(8) and rpc.mountd(8) for further information.
#
# The PARANOID wildcard matches any host whose name does not match its
# address.
#
# You may wish to enable this to ensure any programs that don't
# validate looked up hostnames still leave understandable logs. In past
# versions of Debian this has been the default.
# ALL: PARANOID
ALL:66.116.242.235
```

Figure 3 : Host Active Response verification — malicious IP automatically appended to `/etc/hosts.deny`/

Module 2: Endpoint File Defense (VirusTotal Integration)

- **Monitored Vector:** Real-time File Integrity Monitoring (FIM) tracking file creation or modification.
- **Threat Intelligence:** Queries the VirusTotal API using the calculated SHA256 file hash.
- **Trigger Condition:** File flagged as positive/malicious by threat engines.
- **Automated Remediation:** Generates a Level 12 Alert on the manager and executes a silent deletion (rm -f) of the malicious payload on the endpoint.

```
(jkr@JKR)-[~]
$ ls -la /tmp/eicar.bin
ls: cannot access '/tmp/eicar.bin': No such file or directory
```

Figure 4 : Figure: Automated payload eradication via silent endpoint file deletion (rm -f).

5. Detection Rules & Configuration Specs

Custom Wazuh Rules (/var/ossec/rules/local_rules.xml)

```
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<group name="local,syslog,sshd,abuseipdb,threat_intel,virustotal,urlhaus,dns_web,">

  <!-- Example SSH Rule -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

  <!-- AbuseIPDB Base Integration Rule -->
  <rule id="100100" level="3">
    <field name="integration">custom-abuseipdb</field>
    <description>AbuseIPDB lookup executed.</description>
  </rule>

  <!-- AbuseIPDB High Abuse Score Alert Rule -->
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    <field name="abuseipdb.abuseConfidenceScore" type="pcre2">^([5-9][0-9] | 100)$</field>
    <description>AbuseIPDB: High Abuse Confidence Score detected!</description>
    <group>threat_intel,abuseipdb,</group>
  </rule>

  <!-- VirusTotal High Severity Alert Rule -->
  <rule id="100200" level="12">
    <if_sid>87105</if_sid>
    <description>VirusTotal: Alert - File matches known malware signature!</description>
    <group>virus,malware,threat_intel,virustotal,</group>
  </rule>

</group>
```

4. Operational Workflow

- **Step 1 - Detection:** Agent captures an event (network connection or new file creation) and sends telemetry to the Wazuh Manager.
- **Step 2 - Enrichment:** Wazuh Manager routes the IP or SHA256 hash through custom integration hooks to external APIs (AbuseIPDB / VirusTotal).
- **Step 3 - Rule Evaluation:** Wazuh rules check if the returning score exceeds the severity thresholds:
 - Rule 100101: Triggers when AbuseIPDB returns a confidence score $\geq 50\%$.
 - Rule 100200: Triggers when VirusTotal detects a malicious file match (>0 detections).
- **Step 4 - Active Response:** The manager instructs the local agent active-response engine to execute immediate containment:
 - Network Containment: Triggers `/var/ossec/active-response/bin/host-deny` to automatically append threat IPs to `/etc/hosts.deny`.
 - Payload Eradication: Removes malicious payloads directly from the target endpoint.

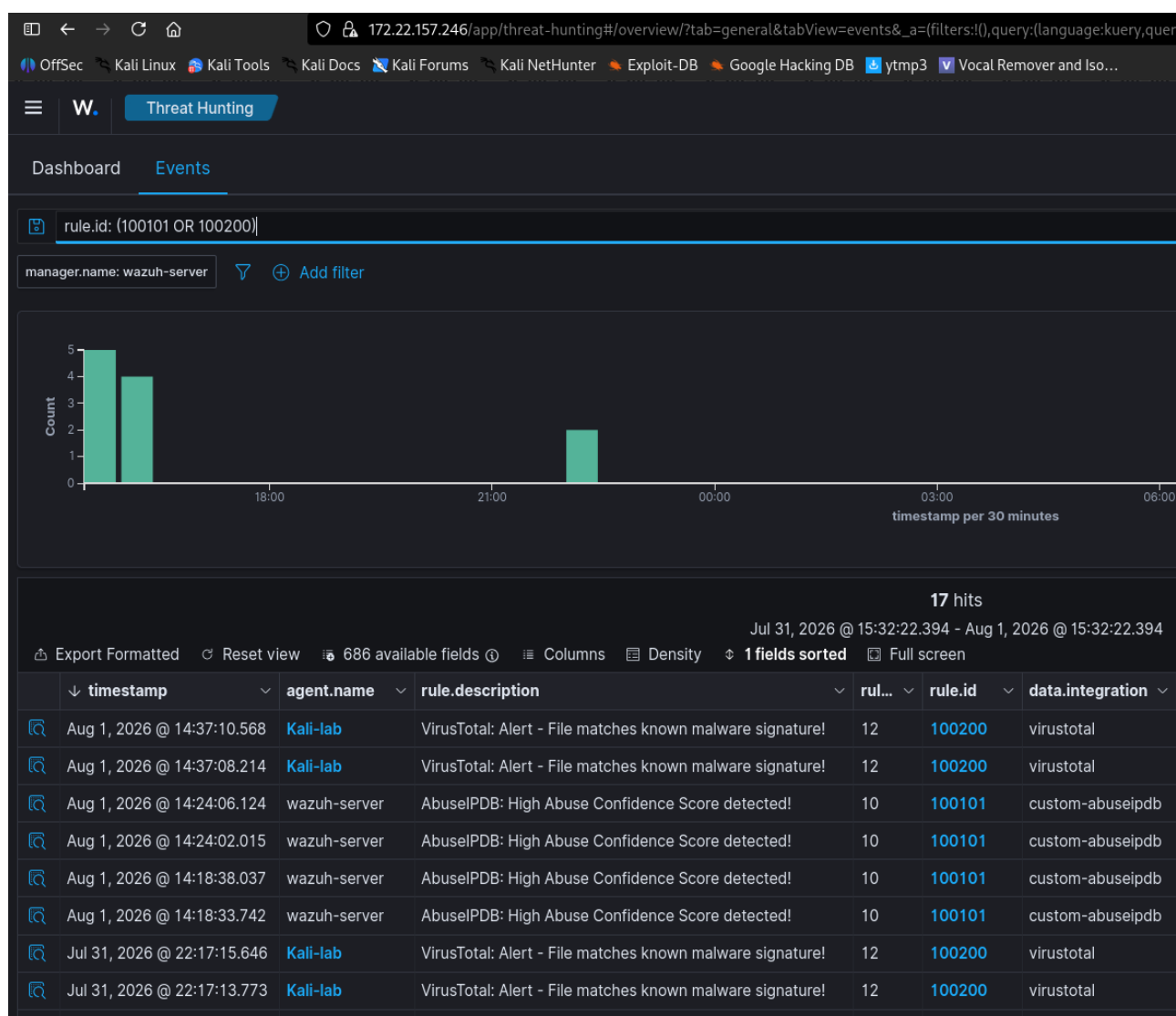


Figure 5: Unified Wazuh SIEM alerts for AbuseIPDB (Rule 100101) and VirusTotal (Rule 100200).

6. Key Benefits

- **Millisecond Reaction Time:** Reduces MTTR from standard human analyst triage averages to automated near-instant execution.
- **Dual-Domain Resilience:** Eliminates blind spots by securing network perimeters and local filesystems simultaneously.
- **Controlled False Positives:** Utilizes strict, calibrated thresholds (e.g., AbuseIPDB score $\geq 50\%$, VirusTotal engine consensus > 0) to avoid disrupting benign operational traffic.
- **Low-Resource Footprint:** Relies on native Wazuh Agent execution daemons without requiring heavy third-party endpoint agents.

7. Technical Requirements

- **SIEM Core:** Active Wazuh Manager (v4.x+) and at least one connected Linux Wazuh Agent.
- **API Credentials:** Valid API keys for AbuseIPDB (v2) and VirusTotal (v2/v3).
- **Operating Systems:** Debian/Ubuntu or RHEL-based Linux distributions.
- **Python Runtime:** Python 3.x installed within the Wazuh framework environment (/var/ossec/framework/python/bin/python3).

8. Future Roadmap

- **URLhaus Integration:** Expand payload enrichment to scan suspicious domain/URL references in web server access logs.
- **AlienVault OTX Pulse Integration:** Add Open Threat Exchange feeds for broader IoC contextual correlation.
- **SOAR Orchestration:** Route Wazuh Active Response webhook alerts through Shuffle SOAR for automated ticket creation in Incident Management tools.